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1. Consider the nonlinear spring-mass-damper system we have been using to demonstrate nonlinear Kalman filters. For this model, the dimension of x_k is two, the dimension of u_k is one, the dimension of z_k is one, the dimension of the process-noise variable w_k is one, and the dimension of the sensor-noise variable v_k is one.

1 / 1 point

How many sigma points are required to model the uncertain random quantities for this model?

9

✔ Correct

Yes, this is correct.

2. Again, for the nonlinear spring-mass-damper system, what is the value of $\alpha_0^{(m)}$ if we use $h = \sqrt{3}$ and the CDKF rules (the same as in the Octave code presented in this lesson)? Enter your answer with two digits to the right of the decimal point. Note that this is the first entry of the variable `wm` in the SPKF initialization code.

1 / 1 point

-0.33

✔ Correct

Yes, that is correct.

3. Again, for the nonlinear spring-mass-damper system, what is the value of $\alpha_1^{(m)}$ if we use $h = \sqrt{3}$ and the CDKF rules (the same as in the Octave code presented in this lesson)? Enter your answer with two digits to the right of the decimal point. Note that this is the second entry of the variable `wm` in the SPKF initialization code.

1 / 1 point

0.17

✔ Correct

Yes, this is correct.